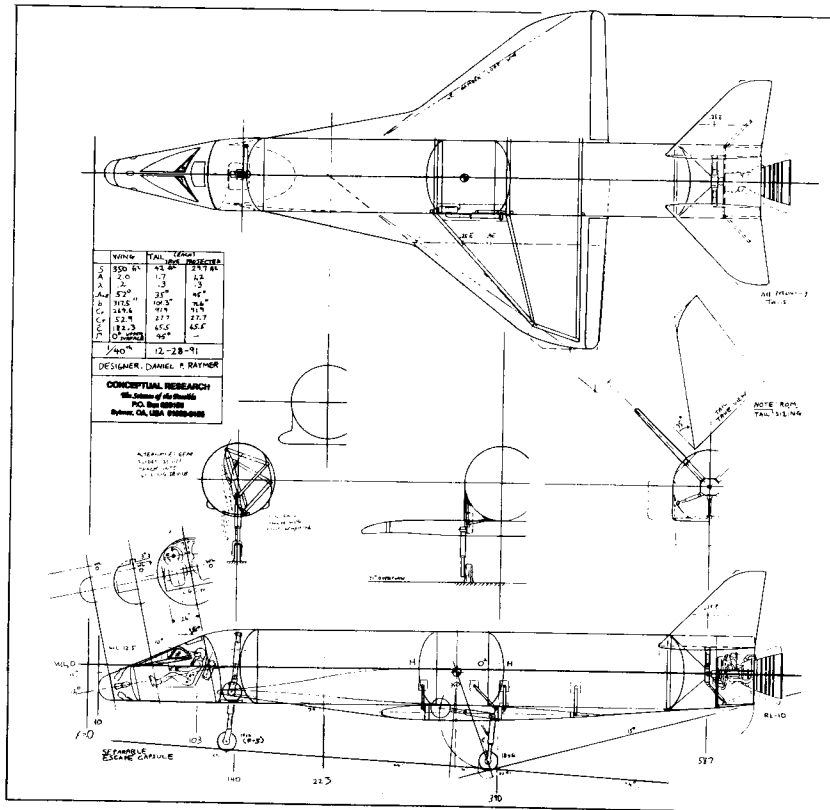


within each cell, using various convergence schemes to equate the flow properties along the boundaries connecting the cells.

While gridding the space around a simple cylinder or a lone wing can be easily automated, the gridding of the flowfield around a full aircraft must currently be done manually and can literally take months. Figure 12.40 (reprinted with permission from Ref. 82) illustrates the complexity of the flowfield gridding. Note, for example, where the canopy meets the fuselage and where the cells must fan out in the empty region between wing and canard.

Gridding is especially important because the CFD results are highly sensitive to the shaping of the cells. You can actually get different answers for the same aircraft using two different gridding schemes. According to the author of Ref. 82, “this sensitivity is more pronounced than that due to the type of mathematical model being used, e.g., NS vs Euler equations.”

To address the gridding problem, researchers are investigating artificial intelligence (AI) approaches to gridding. Another approach is the computationally-adaptive gridding in which the gridding scheme is automatically adjusted based upon the CFD results.



Skyrocket Aerospace Plane Concept

13 PROPULSION

13.1 INTRODUCTION

All forms of aircraft propulsion develop thrust by pushing air (or hot gases) backward. In a simplified case the force obtained can be determined using Newton's equation ($F = ma$) by summing all the accelerations imparted to the air.

This is shown for fluid flow in Eq. (13.1), where s is the cross-sectional area of the fluid acted upon by the propulsion system, V is the fluid velocity, and the zero subscript indicates the freestream condition. The propeller or jet engine is assumed to “magically” accelerate the air from velocity V_0 to V .

The rate of useful work done by the propulsion system, called the “thrust power (P_t),” equals the product of the thrust force and the aircraft velocity [Eq. (13.2)].

The kinetic energy imparted to the fluid by the propulsion system is determined by the difference in fluid velocity, as shown in Eq. (13.3).

The “propulsion efficiency (η_{PE})” is defined as the ratio of thrust power obtained to energy expended, as shown in Eq. (13.4). Note that the efficiency is maximized when there is no change in fluid velocity. Unfortunately, at this condition Eq. (13.1) shows that the thrust is zero!

$$F = ma = \dot{m}\Delta V = (\rho Vs)(V - V_0) = \rho sV(V - V_0) \quad (13.1)$$

$$P_t = FV_0 = \rho sV(V - V_0)V_0 \quad (13.2)$$

$$\Delta E = \frac{1}{2}\dot{m}V^2 - \frac{1}{2}\dot{m}V_0^2 = \frac{1}{2}\rho Vs(V^2 - V_0^2) = \frac{\rho s}{2}V(V^2 - V_0^2) \quad (13.3)$$

$$\eta_{PE} = \frac{P_t}{\Delta E} = \frac{2}{V/V_0 + 1} \quad (13.4)$$

There is an unavoidable tradeoff between thrust and efficiency as determined by the ratio between exhaust and freestream fluid velocity. For maximum thrust, this ratio must be very high. For maximum efficiency, this ratio should be unity.

If the exhaust velocity is reduced to little more than the freestream velocity for improved efficiency, the cross-sectional area of fluid affected by the

propulsion system must approach infinity to maintain constant thrust. A typical turbojet will operate with the ratio of exhaust velocity to freestream velocity at well above 3.0, whereas a typical propeller aircraft will operate with this ratio at about 1.5.

The analysis above is too simplistic for actual thrust calculation. It falsely assumes that the fluid velocity is constant throughout the exhaust and that all of the accelerations experienced by the airmass occur at the propeller plane or within the jet engine.

Actually, the exhaust of a jet engine is usually at a higher pressure than the outside air, so the flow expands after leaving the nozzle. In other words, the air is still accelerating after the aircraft has passed.

For a propeller, the airmass acceleration doesn't even occur at the propeller disk! Roughly half the airmass acceleration occurs before reaching the propeller, and the other half occurs after passing the propeller.

Propulsion force estimation is also complicated by the fact that the propeller flowfield or jet intake and exhaust will influence the whole flowfield of the aircraft. It has already been mentioned that a pusher propeller will reduce the drag of a stubby aft fuselage by "sucking" air inward and preventing flow separation. Should this reduced drag be considered a part of the propulsive force because it is controlled with the throttle? What about the increased drag due to the propeller wake on a conventional airplane?

For a propeller aircraft, most of the propulsive force is exerted directly on the aircraft by the pull (or push) of the propeller itself through the propeller shaft. The propeller shaft is usually connected to the engine so the engine mounts actually pull (or push) the rest of the aircraft through the air.

For a jet aircraft the force exerted through the engine mounts may only be a third of the total propulsive force. Figure 13.1 shows the thrust contributors for a typical Mach 2.2 nacelle. The engine itself only contributes about 8% of the total. The nozzle, which generates thrust by expanding the high-pressure engine exhaust, contributes almost 30%.

The inlet system uses a system of shocks to slow the air to a subsonic speed. This creates a substantial drag. However, the subsonic expansion inside the inlet duct contributes a positive force that more than makes up

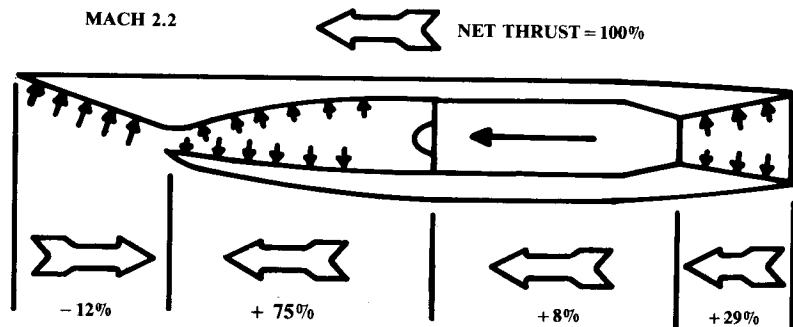


Fig. 13.1 Turbojet thrust contributors.

for the external inlet system drag. In fact, the subsonic expansion inside the inlet duct is the largest single contributor of thrust! This illustrates the difficulty of calculating thrust by any simple model.

This chapter provides methods for estimating the net thrust provided by a propeller or jet engine as a part of the overall vehicle analysis and optimization. These methods are simplified to permit experiencing the whole design process within the time devoted to the typical college design course. The chapter will also introduce the vastly more complicated process of installed-thrust estimation used at major aerospace firms. Reference 46 (highly recommended) provides a detailed treatment of jet-engine design and installation.

13.2 JET-ENGINE THRUST CONSIDERATIONS

Before we begin the discussion of jet-engine installed thrust, a brief introduction to jet-engine cycle analysis and its effect on design is in order. As mentioned in Chapter 10 and shown in Fig. 10.1, a jet engine develops thrust by taking in air, compressing it (via the inlet duct and the compressor), mixing in fuel, burning the mixture, and expanding and accelerating the resulting high-pressure, high-temperature gases out the rear through a nozzle.

To provide power to drive the compressor, a turbine is placed in the exhaust stream which extracts mechanical power from the high-pressure gases. If greater thrust is required for a short period of time, an afterburner can be placed downstream of the turbine permitting the unburned air in the turbine exhaust to combust with additional fuel and thereby increase the exhaust velocity.

"Gross thrust" is produced as a result of the total momentum in the high-velocity exhaust stream. "Net thrust" is calculated as the gross thrust minus the "ram drag," which is the total momentum in the inlet stream. Note that the ram drag, which results from the deceleration of the air taken into the inlet, is included in the engine cycle analysis performed by the engine manufacturer to determine net uninstalled thrust.

Jet-engine cycle analysis, as detailed in Ref. 46 and other propulsion texts, is the straightforward application of the laws of thermodynamics to this Brayton engine cycle. In an "ideal" analysis, the efficiencies of components such as compressors and turbines are assumed to be 100%, i.e., no losses, and the resulting thrust for a given fuel flow, altitude, and Mach number is calculated. While optimistic, such ideal cycle analysis produces results in the "right ballpark" and illustrates the trends produced by varying such cycle parameters as overall pressure ratio, turbine inlet temperature, bypass ratio, and flight condition. These are discussed below.

One over-riding factor in the determination of jet-engine performance is that the net thrust produced is roughly proportional to the air mass flow (velocity $\times$ air density $\times$ airflow cross section) entering the engine. For a modern afterburning turbojet engine, roughly 126 pounds of thrust (the "specific net thrust") is developed for each pound per second of air taken in by the engine. For a turbofan engine, a specific net thrust of roughly 40–60 can be obtained.

An increase in air density such as at low altitude or low outside air temperature would therefore increase thrust by increasing mass flow. Hot day takeoffs from a high-elevation airport such as Denver pose problems because the reduction in air density causes a reduction in mass flow, and hence, thrust.

Similarly, an increase in aircraft velocity also increases thrust due to ram effect increasing the mass flow. However, for a typical subsonic jet, the exhaust comes out the nozzle at a choked condition, and so the exit velocity equals the speed of sound regardless of aircraft velocity. As aircraft velocity approaches the speed of sound, the thrust is therefore reduced for the choked exit nozzle [see Eq. (13.1)]. When combined with the favorable ram effect, this results in a relatively constant thrust as velocity increases for the typical subsonic jet, dropping off as transonic speeds are reached.

For supersonic jet engines, a variable area, converging-diverging nozzle is typically employed which permits supersonic exhaust velocities. Therefore, the ram effect does cause the thrust to tend to increase with increasing velocity until at high Mach numbers where excessive total pressure losses occur in the inlet, resulting in thrust degradation. The Mach number at which inlet losses become excessive is determined by the number of shocks and the extent of variable geometry employed, as described in Chapter 10.

Thrust and propulsive efficiency are strongly affected by the engine's overall pressure ratio (OPR). OPR is the ratio of the pressures at the engine exhaust plane and inlet front face. This pressure ratio is a measure of the engine's ability to accelerate the exhaust, which produces thrust. OPRs usually range from about 15 to 1 to about 30 to 1.

Another key parameter which currently limits turbine engine performance is the turbine inlet temperature (TIT). As mentioned earlier, it would be desirable for maximum thrust and efficiency to combust at the stoichiometric air-fuel ratio of about 15 to 1. This produces temperatures far too high for current turbine materials, even using the best available cooling techniques. Instead, a "lean" mixture of about 60 to 1 (air to fuel) is used, with the extra air holding down the TIT to about 2000–2500°F. This results in less thrust and thermal efficiency, and so a key objective in propulsion technology development has always been the increase in allowable TIT.

To increase propulsive efficiency, the turbofan engine uses an oversized fan with some of the accelerated fan air "bypassed" around the engine, not being used for combustion. This has the effect of allowing the engine to accelerate a larger cross-sectional area of air by a smaller change in velocity, which increases efficiency as determined by Eqs. (13.1) and (13.4). The bypass ratio was defined in Chapter 10 as the ratio of the mass flows of the bypassed air and the air that goes through the core of the engine to be used for combustion.

A higher bypass ratio, which enables the engine to accelerate a larger cross section of air, produces higher efficiency and hence greater thrust for a given expenditure of fuel. However, the fan alone cannot efficiently accelerate the air to transonic or supersonic exit speeds, and so this favorable effect works only at lower speeds. As was shown in Fig. 10.2, the high-bypass turbofan is best at subsonic speeds, giving way to the low-bypass ratio turbofan at the low supersonic speeds. At higher supersonic speeds, say over about Mach 2.2, the pure turbojet is superior.

13.3 TURBOJET INSTALLED THRUST

Chapter 10 described statistical methods for estimating installed thrust and specific fuel consumption for jet engines. These are suitable for initial sizing and performance estimation. For a more sophisticated analysis it is necessary to estimate analytically the installed performance of an existing engine or a proposed new engine from the uninstalled engine data.

Uninstalled engine data can be obtained from the engine manufacturer. Data for several conceptual engines is summarized in the appendices. For a new-design engine, the data must be estimated using cycle-analysis equations such as in Ref. 46. Usually the designers in an aircraft company will obtain the uninstalled-engine data for a proposed engine from the engine manufacturer, and then will correct the data for installation effects.

It is common early in design studies to approximate the performance of a new-design engine by a "fudge-factor" approach. An existing engine with approximately the same bypass ratio is selected, and its size, weight, and performance data are multiplied by factors based upon the expected improvements by applying advanced technologies.

For example, it might be assumed that an engine designed ten years from now would have 25%-less specific fuel consumption, 30%-less length, and 30%-less weight compared to an existing engine. Such "fudge-factors" are based upon either historical trend analysis or an approximate cycle analysis for expected technology improvements.

It is assumed below that uninstalled-engine data is available, either from an engine manufacturer, a preliminary cycle analysis, or a fudge-factor approach based upon some given engine (such as those in the appendices).

13.4 THRUST-DRAG BOOKKEEPING

Bookkeeping is not normally considered an engineering subject. However, the interactions between thrust and drag are so complex that only a bookkeeping-like approach can assure that all forces have been counted once and only once. It is not at all uncommon to discover, halfway through an aircraft design project, that some minor drag item has been either included in both drag and thrust calculations or has been ignored by both departments under the assumption that it is being included by the other!

Each aircraft company develops its own system for thrust-drag bookkeeping. In fact, a different system may be developed for each design project. In most cases the guiding principle for determining whether a force is considered a part of the drag or the thrust is whether that force changes when the throttle setting is changed.

In an afterburning jet engine, for example, the nozzles open wide when the throttle is advanced to the afterburning position. This changes the aerodynamic drag on the outside of the nozzles, so the nozzle aerodynamic drag is counted as a reduction in the engine thrust in many thrust-drag bookkeeping systems.

In other thrust-drag bookkeeping systems, the nozzle drag is separated into two components: (1) the drag value at some fixed nozzle setting (usually full open), which is included in the aerodynamic drag calculation, and

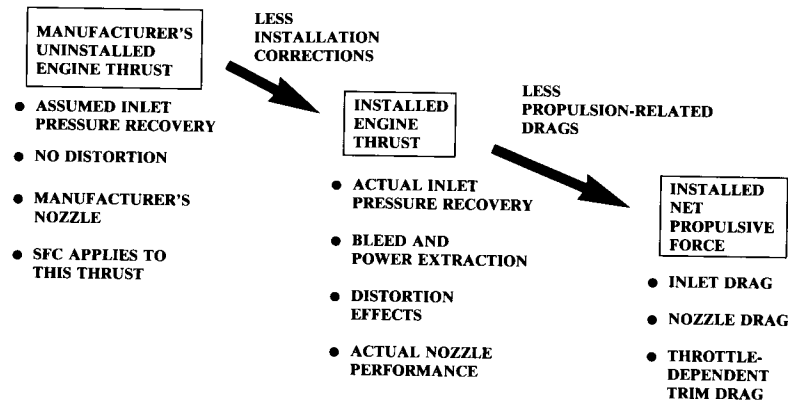


Fig. 13.2 Installed thrust methodology.

(2) the variation of drag as the nozzle setting is changed, which is included in the propulsion-installation calculations.

Either bookkeeping approach will give correct results providing that the Aerodynamics and Propulsion departments both understand it.

Thrust-drag bookkeeping becomes especially complex when sorting out the results of wind-tunnel testing. Different wind-tunnel models are used to test different thrust and drag items. The model used for determining basic aerodynamic and stability derivatives is usually unpowered, and a separate powered model is used to estimate propulsion effects. Lack of a mutually understood bookkeeping system by both the Aerodynamic and Propulsion departments will cause chaos.

The student should realize that the organization of this book assumes a thrust-drag bookkeeping system. Items presented in this chapter as reductions to thrust may be considered to be drag items in another bookkeeping system. Reference 35 contains a detailed review of the subject of thrust-drag bookkeeping.

13.5 INSTALLED-THRUST METHODOLOGY

The actual available thrust used in performance calculations—called the “installed net propulsive force”—is the uninstalled thrust corrected for installation effects, minus the drag contributions that are assigned to the propulsion system by the selected thrust-drag bookkeeping system. This is depicted in Fig. 13.2.

The “manufacturer’s uninstalled engine thrust” is obtained by cycle analysis and/or testing, or can be approximated using fudge-factors as described above. The manufacturer’s engine data is based upon some assumed schedule of inlet pressure recovery vs Mach number. Inlet pressure recovery (P_1/P_0) is the total pressure at the engine front face (1) divided by the total pressure in the freestream (0).

For subsonic engines, it is frequently assumed that the inlet pressure recovery is perfect, i.e., $P_1/P_0 = 1.0$. For supersonic military aircraft a Mil-Spec formula is used. Inlet distortion, engine bleed, and engine power extraction are usually assumed by the manufacturer to be zero. Also, the engine data is based upon the manufacturer’s nozzle design.

Note that the SFC values supplied with the engine are based upon this uninstalled-engine thrust, not the installed net propulsive force. When determining fuel usage, the SFC values must be ratioed up by the ratio between uninstalled-engine thrust and the net propulsive force (i.e., the thrust required for the desired performance).

The “installed-engine thrust” is the actual thrust generated by the engine when installed in the aircraft. This is obtained by correcting the thrust for the actual inlet pressure recovery and nozzle performance, and applying thrust losses to account for engine bleed and power extraction.

“Inlet distortion” refers to pressure and velocity variations in the air delivered to the engine. It primarily affects the allowable operating envelope of the engine.

The installed net propulsive force is the installed engine thrust minus the inlet, nozzle, and throttle-dependent trim drags. The steps depicted in Fig. 13.2 are detailed below.

Installed Engine Thrust Corrections

The manufacturer’s uninstalled engine thrust is based upon an assumed inlet pressure-recovery. For a subsonic engine, it is typically assumed that the inlet has perfect recovery, i.e., 1.0. Supersonic military aircraft engines are usually defined using an inlet pressure-recovery of 1.0 at subsonic speeds and the inlet recovery of Eq. (13.5) (MIL-E-5008B) at supersonic speeds. Added to this is the pressure recovery loss due to internal flow in the inlet duct itself. Typically this adds 2–3% to the losses.

Figure 13.3 shows this reference inlet pressure-recovery plotted vs Mach number, compared to the recovery available for a normal-shock inlet and external compression inlets with one, two, and three ramps.

$$\left(\frac{P_1}{P_0}\right)_{\text{ref}} = 1 - 0.075(M_\infty - 1)^{1.35} \quad (13.5)$$

The external compression inlets of Fig. 13.3 are of movable ramp design with a perfectly optimized schedule of ramp angles as a function of Mach number. To determine the pressure recovery of a fixed or less-than-perfect inlet, the shock tables in the appendix should be used.

The pressure losses inside the inlet duct must also be accounted for. These losses are determined by the length and diameter of the duct, the presence of bends in the duct, and the internal Mach number.

For initial evaluation of a typical inlet duct, an internal pressure recovery of 0.96 for a straight duct and 0.94 for an S duct may be used. The short

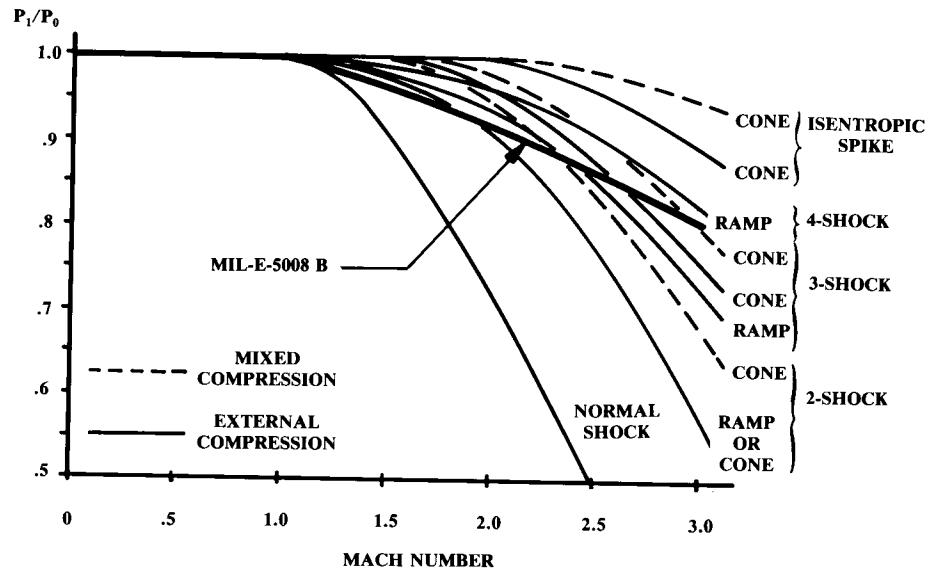


Fig. 13.3 Reference and available inlet pressure recovery.

duct of a subsonic podded nacelle will have a pressure recovery of 0.98 or better. More detailed estimation of inlet internal-pressure loss is based upon experimental data (see Ref. 27), and requires a separate evaluation at each Mach number.

Figure 13.4 provides the actual inlet pressure recoveries of some existing designs. This figure may be used for pressure-recovery estimation during early design studies.

Reducing inlet pressure recovery has a greater-than-proportional effect upon the engine thrust, as shown in Eq. (13.6). The inlet “ram recovery correction factor (C_{ram})” is provided by the manufacturer for various altitudes, Mach numbers, air temperatures, and thrust settings. Typically, C_{ram} ranges from 1.2–1.5. If the manufacturer’s data is not available, C_{ram} may be approximated as 1.35 for subsonic flight and by Eq. (13.7) for supersonic flight.

$$\text{Percent thrust loss} = C_{ram} \left[\left(\frac{P_1}{P_0} \right)_{ref} - \left(\frac{P_1}{P_0} \right)_{actual} \right] \times [100] \quad (13.6)$$

$$\text{Supersonic: } C_{ram} \cong 1.35 - 0.15(M_\infty - 1) \quad (13.7)$$

High-pressure air is bled from the engine compressor for cabin air, anti-icing, and other uses. This engine bleed air (not to be confused with inlet boundary-layer bleed and other forms of secondary airflow) exacts a thrust

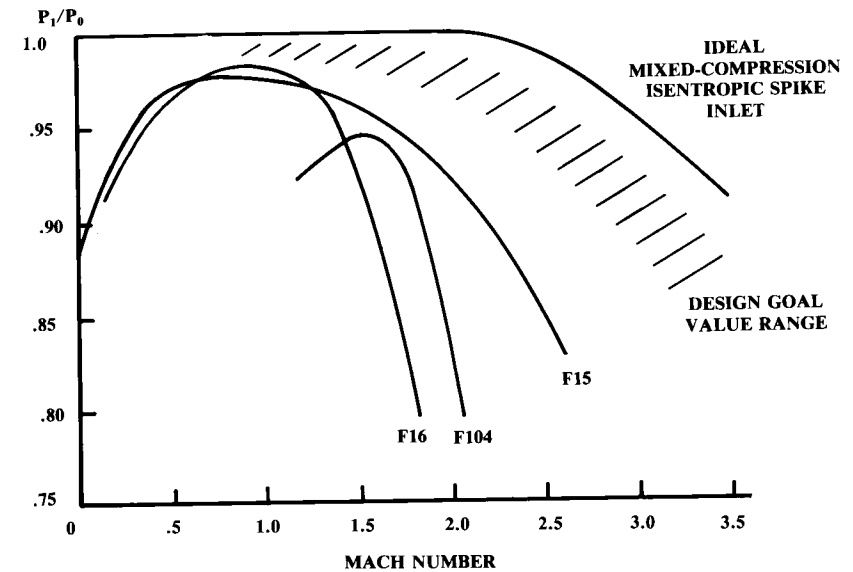


Fig. 13.4 Actual inlet pressure recoveries.

penalty that is also more-than-proportional to the percent of the total engine massflow extracted as bleed air.

Equation (13.8) illustrates this, where the “bleed correction factor (C_{bleed})” is provided by the manufacturer for various flight conditions. For initial analysis, C_{bleed} can be approximated as 2.0. The bleed massflow typically ranges from 1–5% of the engine mass-flow.

$$\text{Percent thrust loss} = C_{bleed} \left(\frac{\text{bleed mass flow}}{\text{engine mass flow}} \right) \times [100] \quad (13.8)$$

Installed engine thrust is also affected by horsepower extraction. Jet engines are equipped with rotating mechanical shafts turned by the turbines. The electrical generators, hydraulic pumps, and other such components connect to these shafts.

This extraction is typically less than 200 hp for a 30,000-lb-thrust engine, and usually has only a small effect upon installed thrust. Horsepower extraction is included in the cycle analysis used for detailed calculation of installed-engine thrust, but can be ignored for initial analysis.

As mentioned, moderate inlet distortion usually has little effect upon installed thrust, but can restrict the engine operating envelope. The effects of distortion are calculated later in the design process. For initial design, the

guidelines previously suggested for location of inlets and for forebody shaping should avoid any later problems with inlet distortion.

Nozzle efficiency has a direct effect upon thrust. However, it is rare to use a nozzle other than that provided by the manufacturer. For cases in which a unique nozzle (such as a 2-D vectoring nozzle) is employed, the new nozzle can usually be designed to provide the same efficiency as the manufacturer's nozzle. (The drag effects of alternate nozzles are discussed later.)

Installed Net Propulsive Force Corrections

The "installed engine thrust" is the actual thrust produced by the engine as installed in the aircraft. However, the engine creates three forms of drag that must be subtracted from the engine thrust to determine the thrust force actually available for propelling the aircraft. This propelling force, the "installed net propulsive force," is the thrust value to be used for aircraft performance calculations.

Most of the engine-related drag is produced by the inlet as a result of a mismatch between the amount of air demanded by the engine and the amount of air that the inlet can supply at a given flight condition. When the inlet is providing exactly the amount of air the engine demands (mass flow ratio equals 1.0), the inlet drag is negligible.

The inlet must be sized to provide enough air at the worst-case condition, when the engine demands a lot of air. This sets the capture area. Most of the time the engine demands less air than an inlet with this capture area would like to provide (i.e., mass flow ratio is less than 1.0).

When the mass flow ratio is less than 1.0, the excess air must either be spilled before the air enters the inlet or bypassed around the engine via a duct that dumps it overboard (Fig. 13.5) or into an ejector-type engine nozzle.

The drag from air spilled before entering the inlet is called "spillage," or "additive" drag. Additive drag represents a loss in momentum of the air which is slowed and compressed by the external part of the inlet but not used by the engine. The additive drag is determined by calculating, for each flight Mach number and engine mass-flow ratio, the Mach numbers and pressures throughout the inlet and integrating the forces in the flight direction for the part of the air which is spilled.

The spilled air will be turned back toward the freestream direction by the inlet cowl lip, producing a reduced pressure on the cowl. This provides a

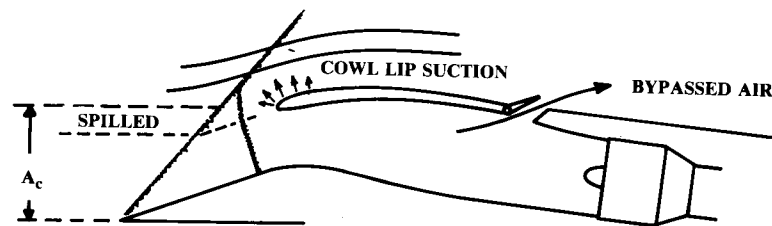


Fig. 13.5 Additive drag, cowl lip suction, and bypass subcritical operation.

suction with a component in the forward direction, i.e., a thrust (as shown in Fig. 13.5). This cowl-lip suction reduces the additive drag by as much as 30–40% in the low-supersonic regime. For a subsonic jet with well-rounded cowl lips, this suction will virtually eliminate additive drag.

Even with cowl-lip suction, the additive drag under certain flight conditions could exceed 20% of the total aircraft drag. A penalty of this magnitude is never seen because the designers resort to inlet-air bypass whenever the additive drag is too great.

Allowing the excess air to enter the inlet and be dumped overboard or into an ejector nozzle, will keep the inlet additive drag to a small value. The resulting bypass drag will be substantially less than the additive drag would have been. Bypass drag is calculated by summing the momentum loss experienced by the bypassed air.

Another form of inlet drag is the momentum loss associated with the inlet boundary-layer bleed. Air is bled through holes or slots on the inlet ramps and within the inlet to prevent shock-induced separation and to prevent the buildup of a thick turbulent boundary layer within the inlet duct. This air is dumped overboard out an aft-facing discharge exit, which is usually located a few feet behind the inlet.

(Note: don't confuse inlet boundary-layer bleed with the inlet boundary-layer diverter. The diverter prevents the fuselage boundary-layer air from entering the inlet. Diverter drag has been accounted for in the aerodynamics chapter.)

Calculation of bleed, bypass, and additive drag including cowl-lip suction is a complicated procedure combining analytical and empirical methods. The textbook methods (see Refs. 26, 27, 25, and 10) are very time-consuming and cannot account for the effects of the actual aircraft geometry, which may greatly affect both the inlet flowfield and the pressure loss through bleed and bypass ducts.

In a major aircraft company such calculations are made by propulsion specialists using complex computer programs. The results are included in the installed net propulsive force data that are provided to the sizing and performance analyst.

To permit rapid initial analysis and trade studies, Fig. 13.6 provides a "ballpark" estimate of inlet drag for a typical supersonic aircraft. This chart was prepared by the author using data from Ref. 47 and other sources, and should be used with great caution as it is merely typical data, not an estimate for any given inlet design.

This chart assumes that the engine is operating at a maximum dry or afterburning power setting, and that the inlet is operating at a corresponding mass-flow ratio. The chart does not reflect the increase in inlet drag experienced when the thrust setting is reduced (which reduces the mass-flow ratio). However, this chart should provide a reasonable approximation of inlet drag suitable for initial analysis and student design studies.

Nozzle drag varies with nozzle position as well as with the flight condition. To properly determine nozzle drag the actual nozzle geometry as a function of throttle setting and flight condition must be known, and the drag calculated by taking into account the overall aircraft flowfield. As an initial approximation, the effect of nozzle position may be ignored and the

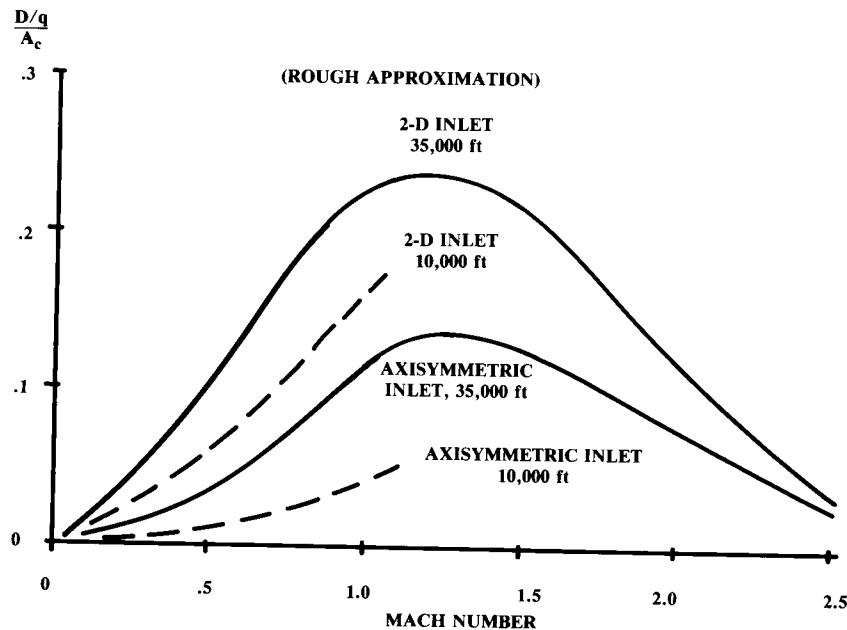


Fig. 13.6 Inlet drag trends.

nozzle drag estimated by the typical subsonic values shown in Table 13.1 (Ref. 10) for the nozzle types shown in Fig. 10.20.

The nozzle drag increases transonically and then drops off at supersonic speeds. For initial analysis the subsonic value may be assumed for all speeds. Note that these nozzle drags are referenced to the maximum cross-sectional area of the fuselage. For a subsonic, podded nacelle, the nozzle drag is negligible.

The remaining propulsion-system drag is the variation of trim drag with throttle setting. If the engine thrust axis is not through the center of gravity, any thrust change will cause a pitching moment. The trim force required to counter this moment is charged to the propulsion in most thrust-drag book-

Table 13.1 Nozzle incremental drag¹⁰

Nozzle type	Subsonic $\Delta C_{D_{Fuselage}}$ *
Convergent	.036-.042
Convergent iris	.001-.020
Ejector	.025-.035
Variable ejector	.010-.020
Translating plug	.015-.020
2-D nozzle	.005-.015

*Referenced to fuselage maximum cross-section area.

keeping systems. For initial analysis this may be ignored unless the thrust line is substantially above or below the aircraft centerline.

13.6 PISTON-ENGINE PERFORMANCE

The aircraft piston engine operates on the four-stroke Otto cycle used by automobiles. The thermodynamic theory of the Otto-cycle reciprocating engine is described in Refs. 28, 48, and 49. For design purposes the most important thing to know about the piston engine is that the horsepower produced is directly proportional to the massflow of the air into the intake manifold. In fact, horsepower is approximately 620 times the air massflow (lb/s).

Mass flow into the engine is affected by the outside air density (altitude, temperature, and humidity) and intake manifold pressure. Equation (13.9) accounts for the air-density effect upon horsepower, and is attributed to Gagg and Ferrar of the Wright Aeronautical Company (1934). This equation indicates that at an altitude of 20,000 ft a piston engine has less than half of its sea-level horsepower.

$$bhp = bhp_{SL} \left(\frac{\rho}{\rho_0} - \frac{1 - \rho/\rho_0}{7.55} \right) \quad (13.9)$$

The intake manifold is usually at atmospheric pressure. A forward-facing air-intake scoop can provide some small increase in manifold pressure at higher speeds. Large increases in manifold pressure require mechanical pumping via a "supercharger" or "turbosupercharger."

The supercharger is a centrifugal air compressor mechanically driven by a shaft from the engine. The amount of air compression available is proportional to engine RPM. The turbosupercharger, or "turbocharger," is driven by a turbine placed in the exhaust pipe. This recovers energy which would otherwise be wasted, and decouples the available amount of compression from the engine RPM.

Supercharging or turbocharging is usually used to maintain sea level pressure in the intake manifold as the aircraft climbs. Typically the sea-level pressure can be maintained up to an altitude of about 15,000-20,000 ft. Above this altitude the manifold pressure, and hence the horsepower, drops. Figure 13.7 shows typical engine performance for nonsupercharged, supercharged, and turbocharged engines.

Supercharging or turbocharging may also be used to raise the intake manifold pressure above the sea-level value to provide additional horsepower from a given engine. However, the increased internal pressures require a heavier engine for structural reasons.

Piston engine performance charts are provided by the manufacturer as a function of manifold pressure, altitude, and RPM.

Propeller Performance

A propeller is a rotating airfoil that generates thrust much as a wing generates lift. Like a wing, the propeller is designed to a particular flight

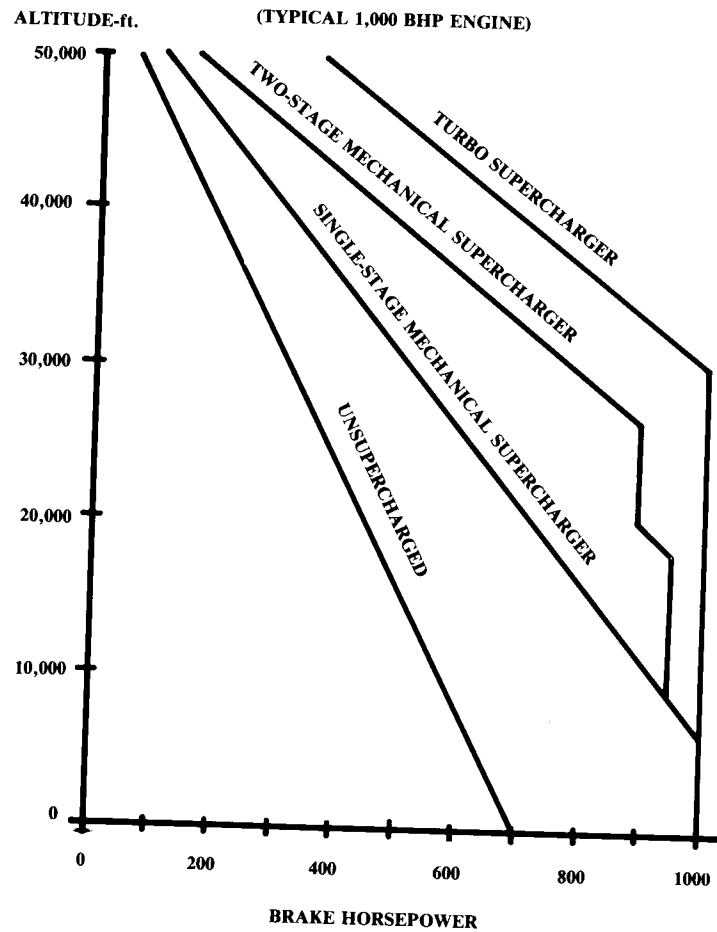


Fig. 13.7 Effects of supercharging.

condition. The propeller airfoil has a selected design lift coefficient (usually around 0.5), and the twist of the airfoil is selected to give the optimal airfoil angle of attack at the design condition.

Since the tangential velocities of the propeller airfoil sections increase with distance from the hub, the airfoils must be set at progressively reduced pitch-angles going from root to tip. The overall “pitch” of a propeller refers to the blade angle at 75% of the radius (70% in some books).

Propeller theory is well covered in many textbooks, such as Ref. 49. While theory is useful for propeller designers, the aircraft designers usually work with experimental propeller data provided by the propeller companies.

As for a wing, the properties of a propeller are expressed in coefficient form. Experimental data for design purposes are expressed using a variety of parameters and coefficients, as described below.

$$\text{Advance Ratio: } J = V/nD \quad (13.10)$$

$$\text{Power Coefficient: } c_P = \frac{P}{\rho n^3 D^5} = \frac{550 \text{ bhp}}{\rho n^3 D^5} \quad (13.11)$$

$$\text{Thrust Coefficient: } c_T = T/\rho n^2 D^4 \quad (13.12)$$

$$\text{Speed-Power Coefficient: } c_S = V \sqrt[5]{\rho/Pn^2} \quad (13.13)$$

Activity Factor:

$$\text{AF}_{\text{per blade}} = \frac{10^5}{D^5} \int_{0.15R}^R cr^3 dr = \frac{10^5 c_{\text{root}}}{16D} [0.25 - (1-\lambda)0.2] \quad (13.14)$$

$$\text{Propeller Efficiency: } \eta_p = \frac{TV}{P} = \frac{TV}{550 \text{ bhp}} = \frac{J c_T}{c_P} \quad (13.15)$$

$$\text{Thrust: } T = \frac{550 \text{ bhp } \eta_p}{V} = \frac{c_T}{c_P} \frac{550 \text{ bhp}}{nD} \quad (13.16)$$

where

- T = thrust (lb)
- V = velocity (ft/s)
- P = power (ft-lb/s)
- bhp = brake horsepower
- n = rotation speed (rev/s)
- D = propeller diameter (ft)
- c = propeller airfoil chord (ft)

The advance ratio (equivalent to the wing angle of attack) is related to the distance the aircraft moves with one turn of the propeller. Advance ratio is sometimes called the “slip function” or “progression factor.”

The power and thrust coefficients are nondimensional measures of those quantities, much like the wing lift-coefficient. The speed-power coefficient is defined as the advance ratio raised to the fifth power divided by the power coefficient. The speed-power coefficient is nondimensional and does not involve the propeller diameter, which is useful for comparison between propellers of different size.

The activity factor is a measure of the amount of power being absorbed by the propeller. Activity factors range from about 90–200, with a typical light-aircraft activity factor being 100 and a typical large turboprop having an activity factor of 140. The final expression in Eq. (13.14) is the activity factor for a straight-tapered propeller blade of taper ratio λ .

Equation (13.15) relates the propeller efficiency, previously discussed in Chapters 3 and 10, to the advance ratio and the ratio of the thrust coefficient to the power coefficient. This ratio is used in Eq. (13.16) to determine the thrust at static conditions when the velocity is zero and the propeller-efficiency equation cannot be used for thrust determination.

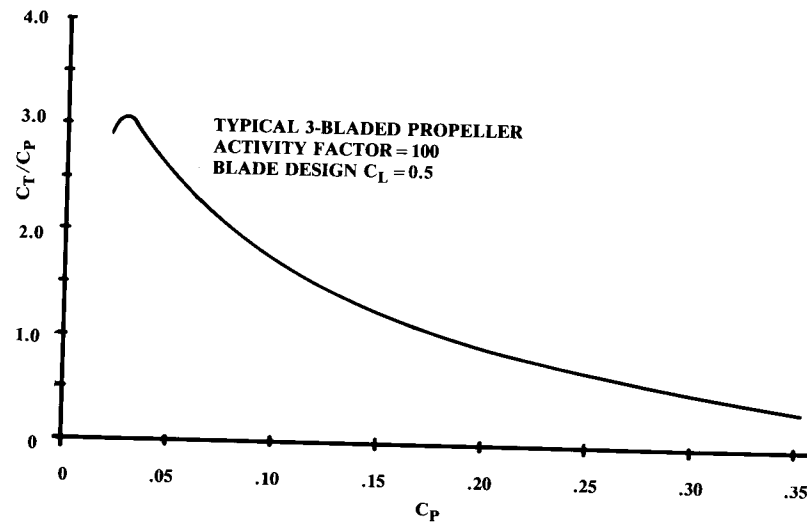


Fig. 13.8 Static propeller thrust. (after Ref. 50)

Propeller data is available from the manufacturers as well as various NASA/NACA reports. This data is provided in many different formats using different combinations of the above parameters and coefficients. Whatever the format, Eq. (13.16) is ultimately used to determine the propeller thrust at a given flight condition.

Figures 13.8 and 13.9, propeller charts for static and forward flight (Ref. 50), have been chosen as typical of propellers used for modern light and business aircraft. These charts represent a three-bladed propeller with a design lift-coefficient of 0.5 and an activity factor of 100.

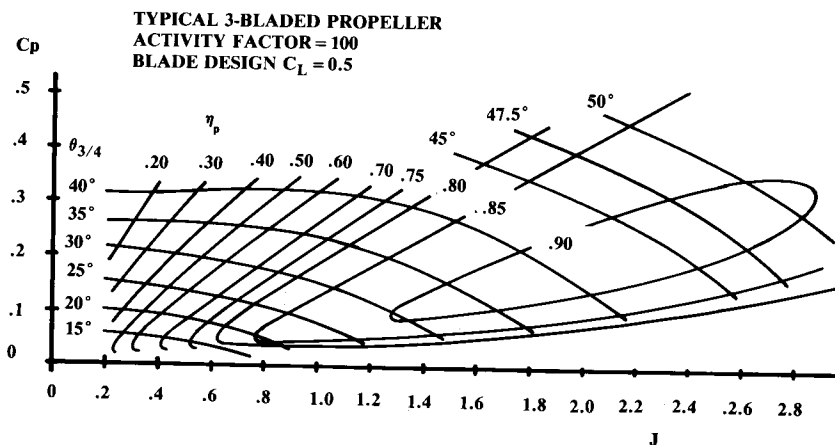


Fig. 13.9 Forward flight thrust and efficiency. (after Ref. 50)

For a two-bladed propeller, the forward-flight efficiencies are about 3% better than shown in Fig. 13.9, but the static thrust is about 5% less than shown in Fig. 13.8. The reverse trends are true for a four-bladed propeller. Also, a wooden propeller has an efficiency about 10% lower due to its greater thickness.

If Ref. 50 is unavailable, Ref. 49 contains 43 pages of propeller charts taken from Ref. 50. Reference 49 also contains a compressibility correction for a propeller with a high tip Mach number (greater than 0.9).

If the propeller is of variable-pitch design, its pitch is adjusted to the optimum blade angle at each flight condition to produce a constant engine RPM regardless of the horsepower being produced.

The advance ratio and power coefficient are then independent variables and the propeller efficiency can be read in Fig. 13.9 for any combination of advance ratio and power coefficient that may occur in flight. Blade angle for the variable-pitch propeller can be read as a fallout parameter in Fig. 13.9.

Remember that the propeller thrust in forward flight is proportional to the inverse of the velocity, which would imply infinite thrust at zero velocity. Instead, the propeller produces the static-thrust value from Fig. 13.8 at zero velocity.

In the speed range from zero to about 50 knots (such as during takeoff), the thrust varies in a fashion that can be represented by a smooth curve faired between the static-thrust value and the calculated forward-flight thrust.

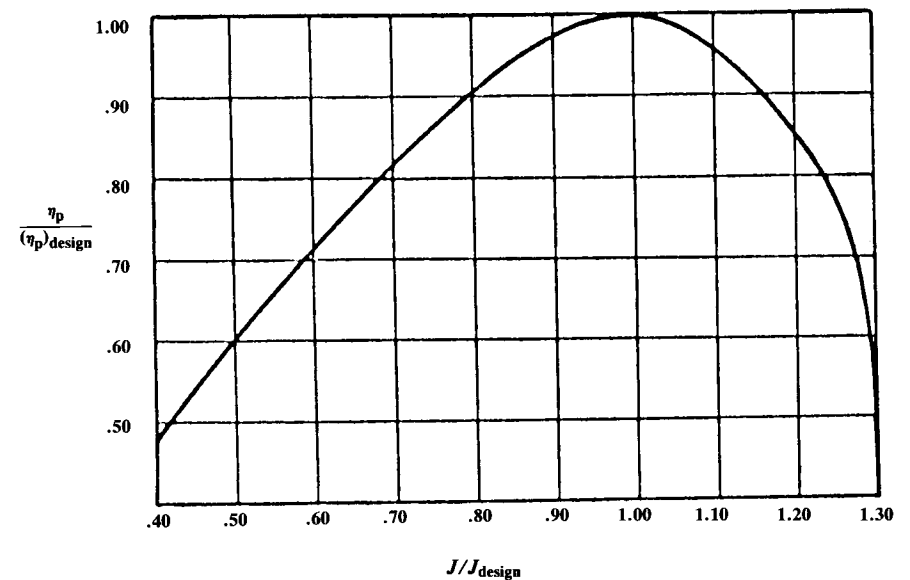


Fig. 13.10 Fixed-pitch propeller adjustment.

If a fixed-pitch propeller is used, the blade angle cannot be varied in flight to maintain engine RPM at any flight condition. Since the RPM and therefore horsepower will vary with velocity, the efficiency and hence the thrust will be reduced at any speed other than the design speed.

Figure 13.9 could be used to determine the thrust from a fixed-pitch propeller by following the appropriate line for the selected blade angle. However, it is simpler to use the approximate method of Fig. 13.10 unless actual propeller data is available.

Figure 13.10 relates the fixed-pitch propeller efficiency at an off-design velocity and RPM to the on-design efficiency, which is attained by the propeller at some selected flight condition. The on-design efficiency is obtained from Fig. 13.9, which is also used to get the required blade angle for the design condition.

The static thrust of a fixed-pitch propeller will be less than is estimated using Fig. 13.8. A fixed-pitch propeller suffers at low speeds due to the high local angles of attack of the blades at low speeds and high RPMs. As a rough approximation it can be assumed that the static thrust is about equal to the thrust at 50 knots for the fixed-pitch propeller.

These charts provide useful rough estimations of propeller performance, but actual charts for the selected propeller should be obtained from the manufacturer for any serious design effort.

Piston-Prop Thrust Corrections

As with jet engines, there are several engine-related drag items that must be considered, namely, scrubbing drag, cooling drag, and engine miscellaneous drag.

Scrubbing drag is the increase in aircraft drag due to the higher velocity and turbulence experienced by the parts of the aircraft within the propwash. This drag could be calculated by determining, for each flight condition, the increased dynamic pressure within the propwash and using that value for the component-drag calculation.

A simpler approach, called the SBAC (Society of British Aircraft Constructors) method, adjusts the propeller efficiency as in Eq. (13.17). The subscript “washed” refers to the parts of the aircraft which lie within the propwash. If the parasite-drag coefficient for the propwashed parts of the aircraft cannot be determined, 0.004 is a reasonable estimate.

$$\eta_{p\text{effective}} = \eta_p \left[1 - \frac{1.558}{D^2} \frac{\rho}{\rho_0} \Sigma(C_{f_e} S_{\text{wet}})_{\text{washed}} \right] \quad (13.17)$$

where C_{f_e} is the equivalent skin-friction (parasite) drag coefficient, referenced to wetted area.

For a pusher-propeller configuration, the scrubbing drag is zero. However, the pusher propeller suffers a loss of efficiency due to the wake of the fuselage and wing. This loss is strongly affected by the actual aircraft configuration, and should equal about 2–5%.

Cooling drag represents the momentum loss of the air passed over the engine for cooling. This is highly dependent upon the detail design of the intake, baffles, and exit.

Miscellaneous engine drag includes the drag of the oil cooler, air intake, exhaust pipes, and other parts. Cooling and miscellaneous drags for a well-designed engine installation can be estimated by Eqs. (13.18) and (13.19) (Ref. 23). However, a typical light aircraft engine installation may experience cooling and miscellaneous drag levels 2–3 times the values estimated by these equations.

$$(D/q)_{\text{cooling}} = (4.9 \times 10^{-7}) \frac{\text{bhp} \cdot T^2}{\sigma V}, \text{ ft}^2 \quad (13.18)$$

$$(D/q)_{\text{misc}} = (2 \times 10^{-4}) \text{ bhp}, \text{ ft}^2 \quad (13.19)$$

where

T = air temperature, deg Rankine

V = velocity in ft/s

13.7 TURBOPROP PERFORMANCE

A turboprop is a jet engine that drives a propeller using a turbine in the exhaust. The jet exhaust retains some thrust capability, and can contribute as much as 20% of the total thrust. For this reason the horsepower rating of a turboprop engine includes the horsepower equivalent of this residual thrust.

This horsepower equivalent of residual thrust is arbitrarily calculated under static conditions as the residual thrust divided by 2.5. Under forward-flight conditions it is calculated using Eq. (13.16) assuming that the propeller efficiency $\eta_p = 0.80$. The total of the mechanical and thrust residual horsepower is called the “equivalent shaft horsepower (ESHP).”

Analysis of the turboprop is a hybrid between the jet and the piston-prop analysis. The engine is analyzed like a jet, including the inlet effects. The residual thrust is provided by the manufacturer as a horsepower equivalent. The propeller is analyzed as described above, including the scrubbing-drag term.

The conventional turboprop, like the piston-prop, is limited by tip Mach number to about Mach 0.7. The turboprop has higher efficiency than the piston-prop at Mach numbers greater than about 0.5 due to the residual jet thrust, but the conventional turboprop is no match for a turbofan engine at the higher subsonic speeds.

Recently, a new type of advanced propeller has been developed that offers good efficiencies up to about Mach 0.85. These are known as “propfans” or “unducted fans (UDF).” They are smaller in diameter than the regular propellers and feature numerous wide, thin, and swept blades. Test programs to date indicate that a well-designed propfan can retain propeller efficiencies of over 0.8 at speeds on the order of Mach 0.85.

Up-to-date data on propfans can be found in publications such as the AIAA’s *Journal of Aircraft* and *Journal of Propulsion and Power*.